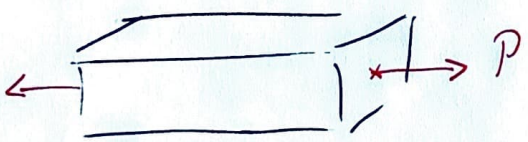
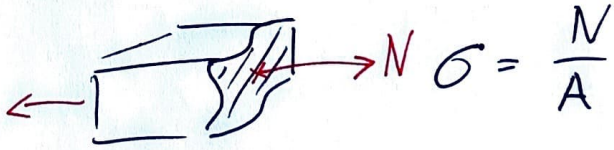
Övning 2 — 2.1: 9, 30, 31 — 2.3: 4, 6

Lösningsväg

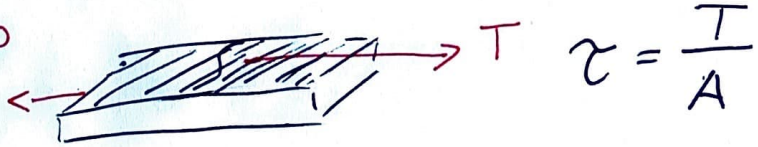
1. Frilägg och snitta
2. Jämvikt
3. Normalspänningar $\sigma = N/A$
4. Töjningar $\varepsilon = \sigma/E$
5. Deformation $\delta = \int_0^L \varepsilon \, dx = \frac{NL}{EA}$ [FS 6.24]

Skjuvning









Statiskt bestämt

nr ekv = nr obk

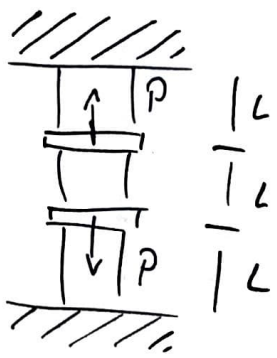
Statiskt obestämt

nr ekv ≠ nr obk

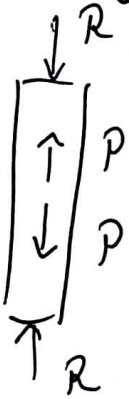
2.1.30

Elastisk modul E

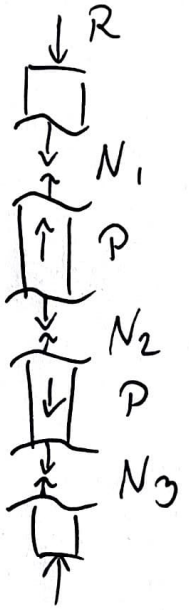
Sökt: Reaktionskraft vid ändstöd, R



① Frilägg



② Snitta



$\underline{N_1 = -R}$

$\underline{N_2 = N_1 + P = P - R}$

$N_2 = N_3 + P = P - R$

$\underline{N_3 = -R}$

4 okända 3 ekv

③ Kompatibilitetssamband

$\delta_{tot} = 0$ $\qquad \delta_{tot} = \delta_1 + \delta_2 + \delta_3 = \sum_{i=1}^{3} \delta_i$

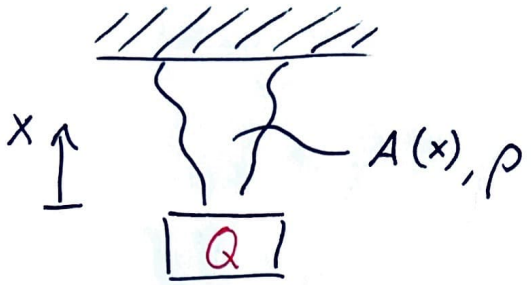
$\delta_i = \varepsilon_i L = \frac{\sigma_i}{E} L = \frac{N_i L}{EA}$

4e ekvation

$\delta_{tot} = \frac{N_1 L}{EA} + \frac{N_2 L}{EA} + \frac{N_3 L}{EA} \Rightarrow \boxed{N_1 + N_2 + N_3 = 0}$

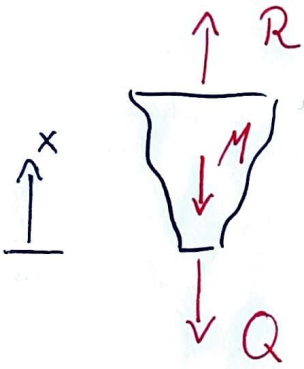
$-R + P - R - R = 0 \Rightarrow \boxed{R = \frac{P}{3}}$

2.1.9

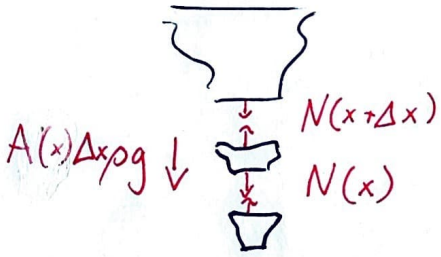


Sökt: Arean $A(x)$ vid max tillåten spänning σ_{till}

① Frilägging



② Snittning



③ Jämvikt

$\downarrow$: $N(x) + A(x)\rho g\Delta x - N(x+\Delta x) = 0$

$$\Rightarrow A(x)\rho g = \frac{N(x+\Delta x) - N(x)}{\Delta x} = \frac{\partial N}{\partial x}$$

Spänningsekvationen $\left(\sigma_{till} = \frac{N(x)}{A(x)}\right)$

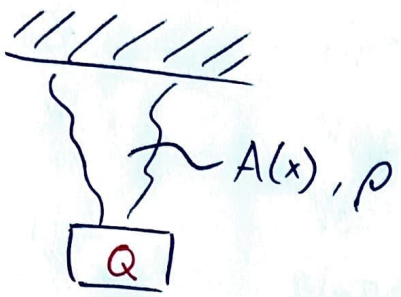
$$\Rightarrow \frac{\partial N}{\partial x} = \sigma_{till}\frac{\partial A}{\partial x}$$

$$\frac{\partial A}{\partial x} = A(x)\rho g / \sigma_{till} \Rightarrow A(x) = C\, e^{\frac{\rho g}{\sigma_{till}}x}$$

Bestäm med randvillkor! $A(0) = \frac{Q}{\sigma_{till}}$

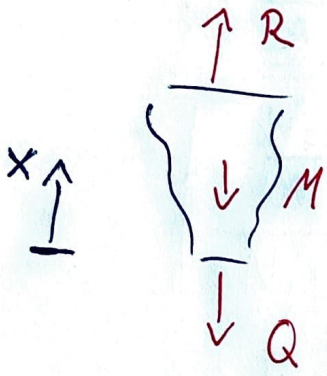
$$\Rightarrow A(x) = \frac{Q}{\sigma_{till}} e^{\frac{\rho g}{\sigma_{till}}x}$$

2.1:9 utan DE

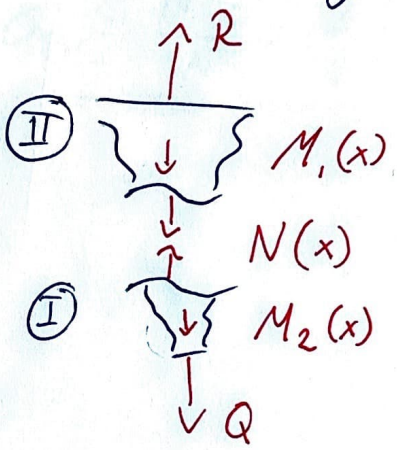


Sökt: Arean $A(x)$ vid max tillåten spänning σ_{till}

① Frilägging



② Snittning



③ Jämvikt

(I) ↑: $N(x) - M_2(x) - Q = 0$

$$N(x) - \rho g \int_0^x A(x)\,dx - Q = 0$$

$$\sigma_{till} A(x) - \rho g \int_0^x A(x)\,dx - Q = 0$$

$$\sigma_{till} \frac{\partial A}{\partial x} - \rho g A(x) = 0$$

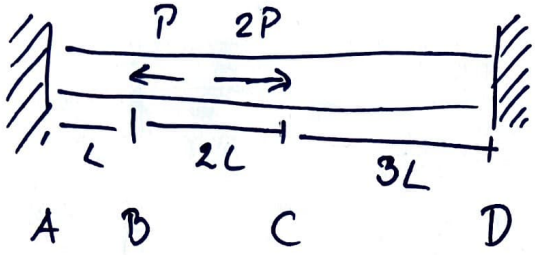
$$\frac{\partial A}{\partial x} = A(x)\frac{\rho g}{\sigma_{till}} \Longrightarrow A(x) = Ce^{\frac{\rho g}{\sigma_{till}}x}$$

Bestäm med randvillkor! $A(0) = \frac{Q}{\sigma_{till}}$

$$A(x) = \frac{Q}{\sigma_{till}} e^{\frac{\rho g}{\sigma_{till}}x}$$

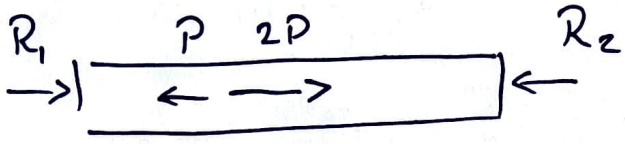
2.1.31

Tvärsnittsarea A
Elasticitetsmodul E



Sökt: Spänningarna i varje del. $\sigma_{AB}, \sigma_{BC}, \sigma_{CD}$

① Frilägging

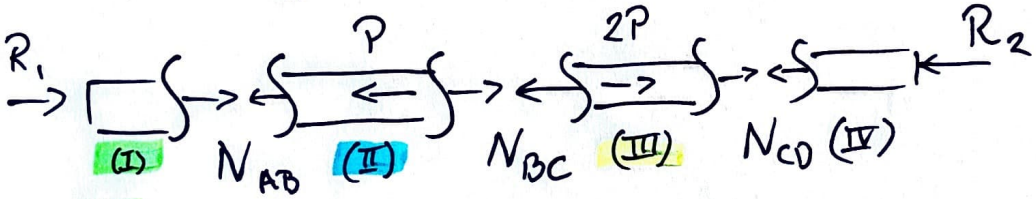


Jämvikt

$\rightarrow: R_1 - P + 2P - R_2 = 0$

$\Rightarrow R_1 = R_2 - P$

② Snittning



(I) $\rightarrow: R_1 + N_{AB} = 0 \Rightarrow N_{AB} = -R_1$

(II) $\rightarrow: N_{BC} - N_{AB} - P = 0 \Rightarrow N_{BC} = N_{AB} + P$

(III) $\rightarrow: N_{CD} - N_{BC} + 2P = 0 \Rightarrow N_{CD} = N_{BC} - 2P$

5 okända 4 ekvationer

③ Kompatibilitetssamband

Inspänt $\Rightarrow \delta_{tot} = 0 \quad \sum_{i=1}^{3} \delta_i = 0$

$\delta_i = \frac{N_i L_i}{EA} \Rightarrow \delta_{AB} = \frac{N_{AB} L}{EA} \quad \delta_{BC} = \frac{N_{BC} 2L}{EA}$

$\delta_{CD} = \frac{N_{CD} 3L}{EA} \Rightarrow N_{AB} + 2N_{BC} + 3N_{CD} = 0$

2.1.31 del 2

④

$$\begin{cases} R_1 = R_2 - P \\ N_{AB} = -R_1 \\ N_{BC} = N_{AB} + P \\ N_{CD} = N_{BC} - 2P \\ N_{AB} + 2N_{BC} + 3N_{CD} = 0 \end{cases} \Rightarrow \begin{cases} N_{AB} = \frac{1}{6}P \\ N_{BC} = \frac{7}{6}P \\ N_{CD} = -\frac{5}{6}P \end{cases}$$

⑤ spänningar

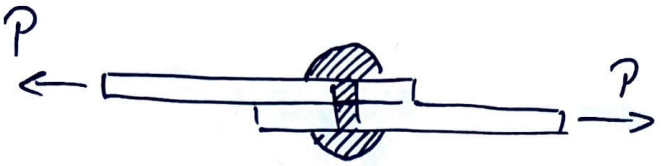
$$\sigma_{AB} = \frac{N_{AB}}{A} = \frac{P}{6A}$$

$$\sigma_{BC} = \frac{N_{BC}}{A} = \frac{7P}{6A}$$

$$\sigma_{CD} = \frac{N_{CD}}{A} = -\frac{5P}{6A}$$

2.3.4

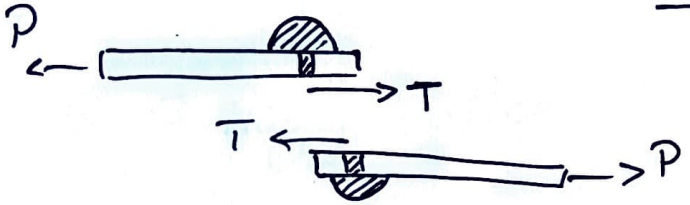
Diameter $d = 25mm$
Kraft $P = 10kN$



Sökt: Genom skiftlig skjuvspänning i niten

τ

② Snittning



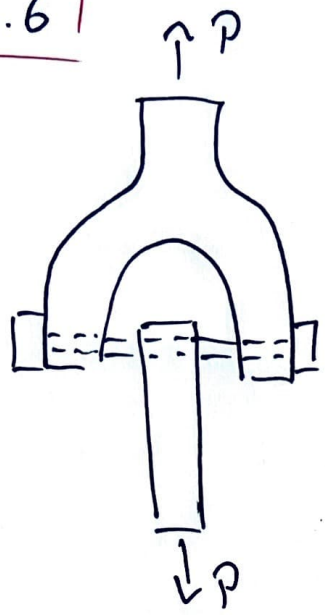
$\rightarrow$: $T - P = 0$ $\quad T = P$

③ Skjuvspänning

$$\tau = \frac{T}{A} = \frac{P}{\frac{d^2\pi}{4}} = \left\{\begin{matrix}\text{insatta} \\ \text{värden}\end{matrix}\right\} = \underline{20,4MPa}$$

$$\boxed{\tau = 20,4MPa}$$

2.3.6

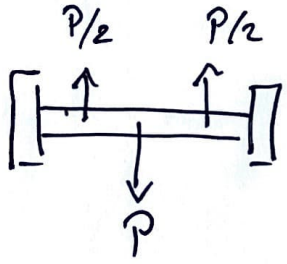


Bultdiameter $d = 10\ mm$

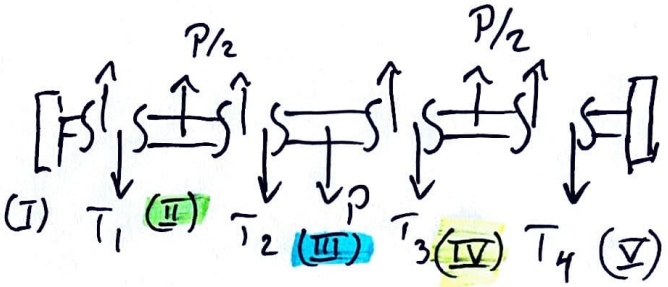
Kraft $P = 1000\ N$

Sökt: Medelskjuvspänningen i bulten. τ

① Frilägg bulten



② Snitta



③ Jämviht

$T_1 = T_4 = 0$

II $\uparrow$: $T_2 + \frac{P}{2} = 0$

III $\uparrow$: $T_3 - T_2 - P = 0$

IV $\uparrow$: $T_3 - \frac{P}{2} = 0$

2 obekant
2 ekv

$\Rightarrow$ $T_2 = -\frac{P}{2}$, $T_3 = \frac{P}{2}$

④ Medel skjuvspänning (tvärsnitt 2 och 3 har samma skjuvkraft fast olika riktning)

$$\tau = \frac{T}{A} = \frac{|T|}{A} = \frac{\frac{P}{2}}{\frac{d^2\pi}{4}} = \frac{2P}{d^2\pi} = 6{,}4\ MPa$$

$\tau = 6.4\ MPa$